Software Studio
軟體設計與實驗

Assignment 01
Web Canvas

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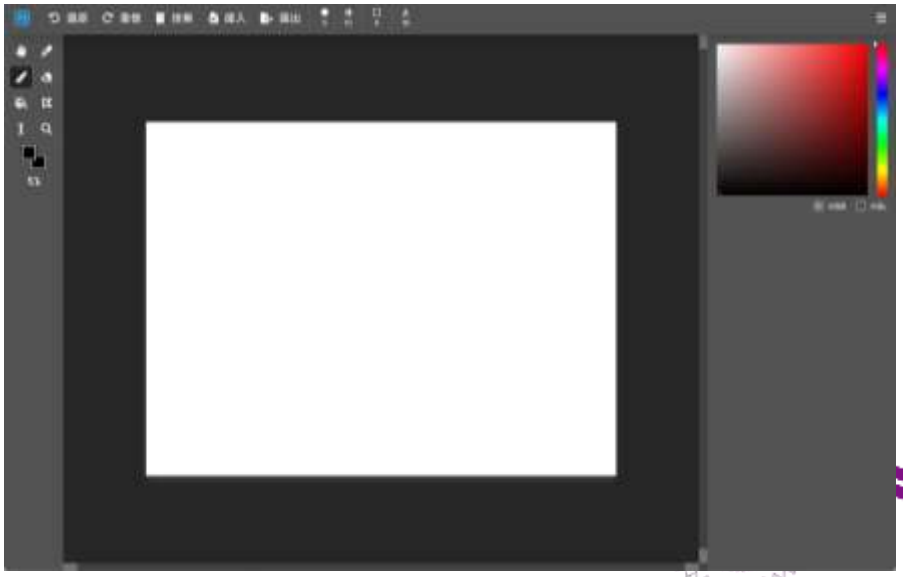
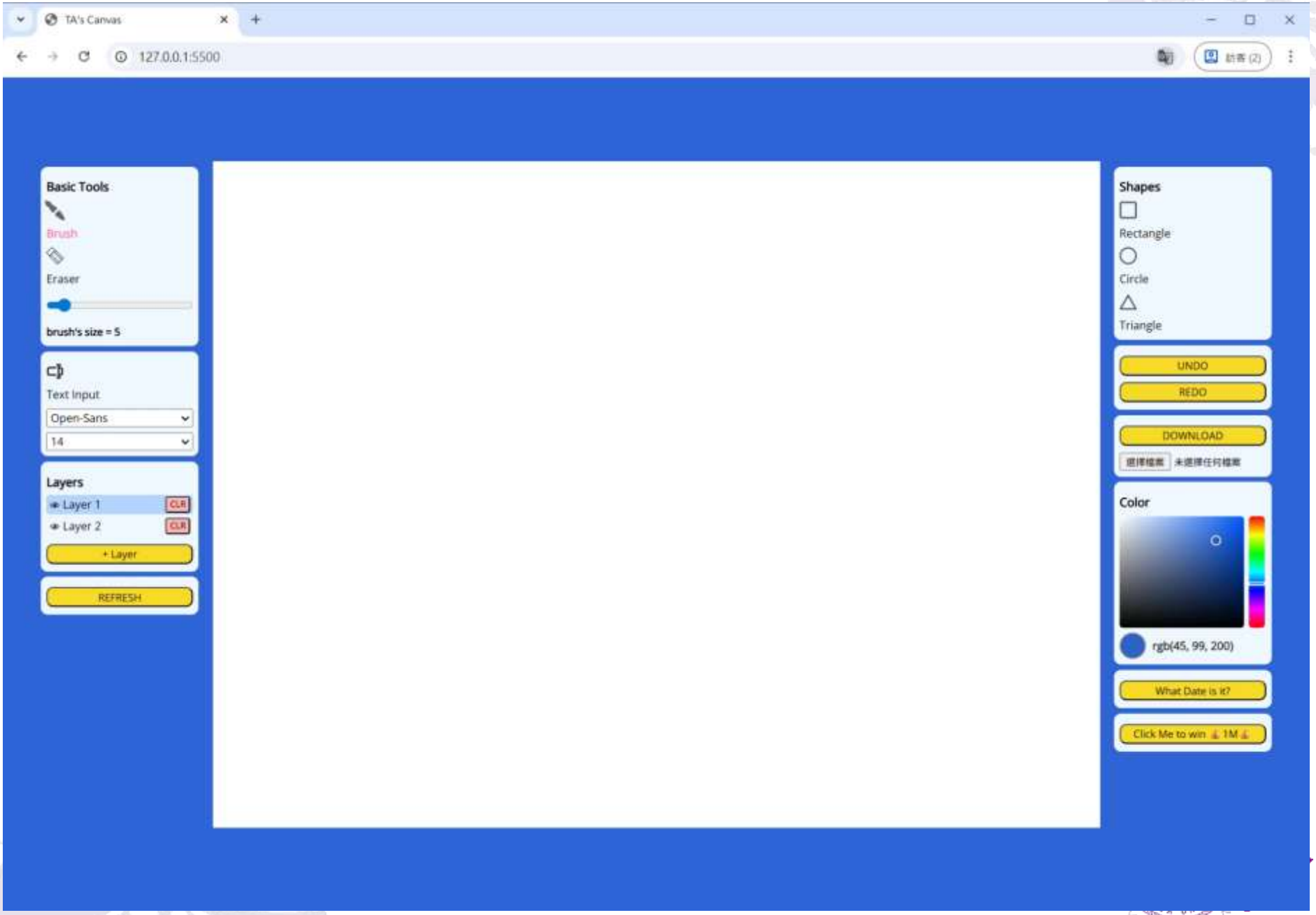
Web Canvas

PENCIL
CUT
ERASER
COPY
LINE
PAST
CIRCLE
INK
UNDO
REDO
RECTANGLE
RANBOW
Width



Goal

- Create your own web page with **HTML5 canvas element** where we can draw somethings.**(HTML)**
- Beautify appearance **(CSS)**.
- Design user interaction widgets and control tools for custom setting or editing **(JavaScript).**
- Describing the functions of your canvas in README.md (you can reference the template by TAs)
- You ought to explain how to operate your web, if TAs can’t find the function, you won’t get the points.


TA's Canvas
127.0.0.1:5500
Basic Tools
Brush
Eraser
brush's size = 5
Text Input
Open-Sans
14
Layers
Layer 1
CLR
Layer 2
CLR
+ Layer
REFRESH
Shapes
Rectangle
Circle
Triangle
UNDO
REDO
DOWNLOAD
Color
rgb(45, 99, 200)
What Date is it?
Click Me to win 1M

Basic components (40%)

- Basic control tools (20%)
  - Brush and Eraser (You can not use white color as eraser!)
  - Color selector (<input type="color"> is not acceptable)
  - Simple menu (brush size, Un/Re-do button)
- Text input (10%)
  - User can click on the canvas and type texts
  - Include a menu to change the typeface (at least 3 options) and font size
- Cursor icon (5%)
  - The cursor icon must change based on the current selected tool
- Refresh button (5%)
  - Reset canvas

Advanced tools (40%)

- Different shape Tools (15%)
  - Circle, rectangle and triangle (-5% for each shape)
  - Drag the cursor to make circles, rectangles, and triangles of different size (Please refer to the example)
  - Flickering brush/canvas is not acceptable
- Image Tool (10%)
  - User can upload image and paste it
  - User can adjust image size afterward
- Download (5%)
  - Download current canvas as an image file
  - The background should not be transparent
- Layer Management (10%):
  - Users can add at least 2 separate layers and toggle layer visibility (Show/Hide)
  - Active Layer: Select which layer is currently being drawn on
  - Clear Layer: A button to reset only the currently selected layer
  - When downloading the image, all visible layers must be composited into a single file

Appearance (10%)

- Interface Design (UI)
  - The application should look professional and modern
  - Pay attention to color schemes, spacing (padding/margin), and consistent button styles
- Usability (UX)
  - Provide clear visual feedback (e.g., highlighting the active tool, hover effects on buttons)
  - Ensure intuitive navigation—users should not be confused about how to change colors or switch layers
- Layout & Responsiveness
  - Use Flexbox or CSS Grid to ensure the layout is organized
  - The canvas should remain accessible and usable on different screen resolutions

Bonus (10%)

- Any Creative Extra Tools (Please describe in README.md)
- Example:
  - Bucket Tool: Filling a closed area with a color
  - Advanced Brush Effects: Spray paint, calligraphy pens…
  - Eyedropper Tool: Click on any pixel on the canvas to "pick" that color
  - Canvas Resizer: Allow users to change the canvas dimensions (Width/Height) via a menu
  - Canvas Rotation: Rotate the entire canvas content by 90 degrees clockwise

Scoring

Item	Score
Basic Tools	40%
Advance Tools	40%
Appearance	10%
Bonus (describe on README.md)	10%

Rules

- Cannot import any third-party code or tools (i.e., color-picker.js)
- No plagiarism
- Write an **AI document** in your project (refer to the details in next page)
- If you have any questions, please ask TA in **eeclass Discussion Forum.**

AI Usage

If you utilize AI tools during development, you must include a report titled **AI_reference.pdf** in the **root directory** of your project. The report must include:

1. **AI Tool(s) Used**: Specify the model (e.g., ChatGPT, Gemini…)
2. **Scope of Usage**: For every segment of code generated or assisted by AI, provide:
   - Location: File name and specific line numbers (e.g., App.js, lines 45-82).
   - Prompt & Response: The exact prompt you used and the AI's output (Screenshots are highly recommended).
   - Refinement & Explanation: Show your modified version of the code and provide a brief explanation of why you made those changes and how the logic works.
3. **Statement of Non-Usage**: If you did not use any AI tools, simply state: "No AI tools were used in this assignment." inside the PDF.

We strongly suggest that you thoroughly analyze any code produced by AI and rewrite it using your own logic, in case that it may be flagged as plagiarism regardless of the documentation.

Reminder

- Deploy your web page to **Firebase page**, and ensure it works correctly. Add TA's account to your firebase project.
  - **Your main page should be named as "index.html"**
- Upload all source code to FTP. (reference eeclass)
  - Compress all files into Assignment01_StudentID.**zip** then upload:
    - Assignment01_StudentID.zip
      - /Assignment01_StudentID
        - index.html, style.css, app.js
        - README.md
        - AI_reference.pdf
        - other source files (if any)
    - if you didn't follow the structure → **-10%**
  - If you upload the files again, please rename to **Assignment01_StudentID_v?.zip (v? -> which version)**
- **MD5 checksum** (if you didn't do this → **-10%**)
- FIRM deadline: 2026/04/09 23:59 (commit time)
- Upload your MD5 and web link to eeclass

作業規則

- 程式碼嚴禁抄襲，抓到抄襲者，抄襲與被抄襲者

直接當掉！直接當掉！直接當掉！

直接當掉！直接當掉！直接當掉！

...

- 抄襲者定義：採用現有資源(線上下載、同學(or 學長姐)的 code)且不經修改直接繳交作業者
- 沒有屍體分數，遲交零分且不得補交或遲交。

如何不被判定抄襲？

- 參考相關網站的程式碼、透過生成式AI產生的程式碼，會因為來源相同、prompt相近導致被Moss判定為抄襲（例如function名稱相同）。
- 因此建議先理解網站\生成式AI\好友討論所產出的程式碼，再用自己的邏輯重寫，以降低被判定為抄襲的可能性。
- 若已知自己有涉及上述的手段來寫作業，建議先用MOSS比對自己的程式碼與參考的資料來源的相似度，確保比例低於20%，以降低被判定為抄襲的可能性。
- 關於Moss的申請與使用，可以參考下面的連結。https://hackmd.io/@GjP3NF8qT0SXa2TjRxSKeQ/HJvqIGerE

作業繳交規則

- 請**務必**透過 MD5 獲得作業checksum 後填入 google 表單。
- 如遇各種原因無法在作業期限前完成上傳，我們將比對 checksum。若 checksum 一致則不算遲交。
- 繳交作業以MD5為主，若有多個 checksum 則取時間最晚的為主。
- MD5 online generator
- MD5 checksum 登記表單
- MD5 checksum 登記查看
- MD5 使用方法

作業繳交流程

請務必遵守以下規則：

1. 將作業打包壓成zip檔，檔名格式以及打包內容以當次作業公告為主。
2. 幫zip檔產生MD5，並填寫google表單。
3. 將zip檔上傳ftp。
4. 將MD5和網址繳交至eeclass。

若違反任何一項繳交作業SOP，
一律扣作業總分10分！！！

貼心提醒：請同學守護zip檔至作業成績公告或期末。